Every term in the sum gives the same contribution, because we integrate over all of them, so we can just fix $j = s+1$, and everything can expressed in the following way

$$\left(\frac{\partial t}{\partial} + \hat{L}_s\right) f_s(1,\dots,s;t) = -\frac{n!}{(n-s)!}(n-s)\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}\dots\mathrm{d}\mu_n\, \hat{P}_{i,s+1}\rho(1,\dots,n;t) =$$

$$= -\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}\hat{P}_{i,s+1}\underbrace{\frac{n!}{(n-s-1)!}\int \mathrm{d}\mu_{s+2}\dots\mathrm{d}\mu_n\rho(1,\dots,n;t)}_{f_{s+1}(1,\dots,s+1;t)},$$

and therefore, compactly

$$\left(\frac{\partial t}{\partial} + \hat{L}_s\right) f_s(1,\dots,s;t) = -\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}\hat{P}_{i,s+1}f_{s+2}(1,\dots,s+1;t) =$$

$$= -\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}\vec{\mathbf{K}}_{i,s+1}\left(\frac{\partial}{\partial p_i} - \frac{\partial}{\partial p_{s+1}}\right) f_{s+1}(1,\dots,s+1;t),$$

and we note that the term $\partial/\partial p_{s+1}$ can be removed (its contribution is zero, and it can be shown through an integration by parts).

From this relation we see that the evolution of f_s depends directly on f_{s+1}.

7 Lesson 7

7.1 Again

The first equation of the hierarchy are the following

$$\left(\frac{\partial}{\partial t} + \frac{1}{m}p_1\frac{\partial}{\partial q_1}\right) f_1(1) = -\int_{r_0}\mathrm{d}\mu_2\vec{\mathbf{K}}_{12}\frac{\partial p_1}{\partial} f_2(1,2)$$

where r_0 limits the radius of interaction,[1] and

$$\left(\frac{\partial}{\partial t}\frac{1}{m}\left(p_1\frac{\partial}{\partial q_1} + p_2\frac{\partial}{\partial q_2}\right)\right) + \vec{\mathbf{K}}_{12}\left(\frac{\partial}{\partial p_1} - \frac{\partial}{\partial p_2}\right) f_2(1,2) = -\int \mathrm{d}\mu_3\left(\vec{\mathbf{K}}_{13}\frac{\partial}{\partial p_1} + \vec{\mathbf{K}}_{23}\frac{\partial}{\partial 2}\right) f_3(1,2).$$

In general, the typical time-scales of these interactions are: the collision time τ

$$\vec{\mathbf{K}}\frac{\partial}{\partial\vec{\mathbf{p}}} \sim \frac{1}{\tau},$$

and the diffusion time τ_s

$$\frac{\vec{\mathbf{p}}}{m}\frac{\partial}{\partial\vec{\mathbf{q}}} \sim \frac{1}{\tau_s}$$

such that the particle travels an amount of space over which f changes significantly. Typically, we have that $\tau \ll \tau_s$.

[1] There has to be another particle in the volume comprised in r_0 for an interaction to occur: the probability of having an interaction is proportional to nr_0^3 where n is the particle density in the gas. In a dilute gas n is very small (the volume per particle is very big).

Usually, to simplify our problem we neglect the integral over μ_3 of f_3: because it is depressed by the density factor, while the other are not.[2] This assumption is equivalent to: it is improbable to have three particles in the same volume r_0^3. Furthermore, f_2 varies (in both time and space) more quickly than f_1: always because its evolution is suppressed by this density factor. Because of this, it is reasonable to assume that f_2 reaches equilibrium faster than f_1.

Correlations between particles in f_2 are due to collisions: for distances $|q_1 - q_2| > r_0$, we can assume that

$$f_2(1,2) = f_1(1)f_1(2),$$

and this is known as the *molecular chaos* assumption. The problem is that in the first equation f_2 is evaluated exactly where this assumption is certainly not true: the region of interaction.

To better understand the problem it is useful to change variables into the frame of reference of the centre of mass. Thus

$$\begin{cases} \vec{\mathbf{R}} = \dfrac{\vec{\mathbf{q}}_1 + \vec{\mathbf{q}}_2}{2} \\ \vec{\mathbf{r}} = \vec{\mathbf{q}}_2 - \vec{\mathbf{q}}_1 \end{cases} \qquad \begin{cases} \vec{\mathbf{P}} = \vec{\mathbf{p}}_1 + \vec{\mathbf{p}}_2 \\ \vec{\mathbf{p}} = \vec{\mathbf{p}}_2 - \vec{\mathbf{p}}_1 \end{cases}$$

and we are also supposing that the force depends only on the distance between the particles

$$\vec{\mathbf{K}}_{21} = -\vec{\mathbf{K}}_{12} = \vec{\mathbf{K}}(\vec{\mathbf{r}}).$$

It is easy to check that Exercise!

$$\vec{\mathbf{p}}_1 \frac{\partial}{\partial q_1} + \vec{\mathbf{p}}_2 \frac{\partial}{\partial q_2} = \ldots = \frac{\vec{\mathbf{P}}}{2} \frac{\partial}{\partial \vec{\mathbf{R}}} + \vec{\mathbf{p}} \frac{\partial}{\partial \vec{\mathbf{r}}}$$

and moreover

$$\frac{\partial}{\partial \vec{\mathbf{p}}_1} - \frac{\partial}{\partial \vec{\mathbf{p}}_1} = \ldots = -\frac{\partial}{\partial \vec{\mathbf{p}}}.$$

If we apply this change of variables, the second equation becomes

$$\left[\frac{\partial}{\partial t} + \frac{\vec{\mathbf{P}}}{2m} \frac{\partial}{\partial \vec{\mathbf{R}}} + \frac{\vec{\mathbf{p}}}{m} \frac{\partial}{\partial \vec{\mathbf{r}}} + \vec{\mathbf{K}}(\vec{\mathbf{r}}) \frac{\partial}{\partial \vec{\mathbf{p}}} \right] f_2 = 0,$$

and this means that if we measure f_2 before and after an infinitesimal time displacement $\mathrm{d}t$

$$f_2(\vec{\mathbf{p}} + \vec{\mathbf{K}}\mathrm{d}t, \vec{\mathbf{r}} + \frac{\vec{\mathbf{p}}}{m}\mathrm{d}ti, t + \mathrm{d}t) = f_2(\vec{\mathbf{p}}, \vec{\mathbf{r}}, t)$$

in the centre of mass frame, by also setting $\vec{\mathbf{P}} = 0$ and $\vec{\mathbf{R}} = 0$, and this determines the classical trajectory of a particle subject to the force $\vec{\mathbf{K}}(\vec{\mathbf{r}})$.

In the stationary equilibrium regime it must be that $\partial f_2/\partial t = 0$, so it follows that

$$\vec{\mathbf{K}}_{12} \left(\frac{\partial}{\partial p_1} - \frac{\partial}{\partial p_2} \right) f_2 = -\frac{1}{m} \left(p_1 \frac{\partial}{\partial q_1} + p_2 \frac{\partial}{\partial p_2} \right),$$

which tells us that the forces on f_2 determine the time-evolution of f_2.

At this stationary regime, we can imagine a constant flux of particles who get deviated by the localised potentials that leaves f_2 always unchanged. A particle can be seen as a peak of f_2 over phase space.

Outside of the interaction region we know that the density distributions are independent, but at the boundaries the are correlated: given $\vec{\mathbf{p}}$ and b (the impact parameter of a particle incoming on a potential), they are correlated with $\vec{\mathbf{p}}'$ of the outgoing particle with the diffusion angle corresponding to the classical trajectory.

From this

Boh

$$\vec{\mathbf{K}}_{12}\left(\frac{\partial}{\partial p_1}-\frac{\partial}{\partial p_2}\right)f_2=-\frac{1}{m}\left(\vec{\mathbf{p}}_1\frac{\partial}{\partial q_1}+p_2\frac{\partial}{\partial q_2}\right)f_2$$

and we can plug this into the equation for f_1: there was a collision term, which contained

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}=-\int_{r_0}\mathrm{d}\mu_2\vec{\mathbf{K}}_{12}\frac{\partial}{\partial p_1}f_2(1,2)=-\int_{r_0}\mathrm{d}\mu_2\vec{\mathbf{K}}_{12}\left(\frac{\partial}{\partial p_1}-\frac{\partial}{\partial p_2}\right)f_2(1,2)$$

where the contribution of the extra term gives zero because of boundary conditions (integration by parts), and thus

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}=\frac{1}{m}\int\mathrm{d}\mu_2\left(p_1\frac{\partial}{\partial q_1}+p_2\frac{\partial}{\partial q_2}\right)f_2=\frac{1}{m}\int\mathrm{d}\mu_2\left(\frac{\vec{\mathbf{P}}}{2}\frac{\partial}{\partial\vec{\mathbf{R}}}+\vec{\mathbf{p}}\frac{\partial}{\partial\vec{\mathbf{r}}}\right)f_2.$$

After the collision $\vec{\mathbf{q}}_1\sim\vec{\mathbf{q}}_2$, thus $\vec{\mathbf{R}}$ does not vary: at the end of the day, it is like if the collision happened in a point. Then

f_2 does not vary with R.

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}=\frac{1}{m}\int\mathrm{d}^3p_2\int_{r<r_0}\mathrm{d}^3q_2\vec{\mathbf{p}}\cdot\frac{\partial f_2}{\partial\vec{\mathbf{r}}}$$

where $\mathrm{d}^3r=b\mathrm{d}b\mathrm{d}\phi\mathrm{d}x$, thus

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}=\frac{1}{m}\int\mathrm{d}p_2|\vec{\mathbf{p}}_2-\vec{\mathbf{p}}_1|\int\mathrm{d}\phi\mathrm{d}b\int_{x_1}^{x_2}\mathrm{d}x\,\frac{\partial f_2}{\partial x}=\frac{1}{m}\int\mathrm{d}p_2|\vec{\mathbf{p}}_2-\vec{\mathbf{p}}_1|\int\mathrm{d}\phi\mathrm{d}b\,b[f_2(x_2)-f_2(x_1)]$$

and this is the trick: we brought f_2 to the boundary, where the chaos hypothesis can be applied, and therefore[3]

$$f_2(x_1)\to f_1(\vec{\mathbf{p}}_1)f_1(\vec{\mathbf{p}}_2)$$

and

$$f_2(x_2)\to f_1(\vec{\mathbf{p_1}}')f_1(\vec{\mathbf{p_2}}')$$

where these primed momenta are the momenta after a scattering with such initial conditions.

In a classical scattering, a cross-section is defined from the following equation

$$N=I\frac{\sigma}{\Omega}\mathrm{d}\Omega$$

where I is the incoming flux of particles, and N is the number of particles (in a unit of time) diffused into a $\mathrm{d}\Omega$ of the direction θ. Classically the outgoing direction θ is determined by the impact parameter b: the particles deflected in a certain $\mathrm{d}\Omega$ around θ are those that enter the process through a surface element (perpendicular to the flux) $b\mathrm{d}b\mathrm{d}\phi$. This means that

$$I\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\mathrm{d}\Omega=Ib\mathrm{d}b\mathrm{d}\phi\implies b\mathrm{d}b\mathrm{d}\phi=\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\mathrm{d}\Omega,$$

which lets us reinterpret our collision term.

Finally

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}=\int\mathrm{d}^3p_2\mathrm{d}\Omega|\vec{\mathbf{v}}_2-\vec{\mathbf{v}}_1|\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\left[f_1(\vec{\mathbf{p_1}}')f_1(\vec{\mathbf{p_2}}')-f_1(\vec{\mathbf{p}}_1)f_2(\vec{\mathbf{p}}_2)\right],$$

and this is called the Boltzmann transport equation, in which only f_1 appears: from now on $f_1\equiv f$.

[2] All the terms are weighted by the inverse of the collision time, so that contribution is net zero.

[3] Technically f_1 depends on the position too, but $x_1\sim x_2$ from the point of view of f_1.